

## **Subject Description Form**

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| <b>Subject Code</b>                                   | EEE2003                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Subject Title</b>                                  | Transportation Engineering Fundamentals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Credit Value</b>                                   | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Level</b>                                          | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Pre-requisite/<br/>Co-requisite/<br/>Exclusion</b> | Nil                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Objectives</b>                                     | <ol style="list-style-type: none"> <li>1. To introduce the fundamental topics and concepts of transportation engineering, including traffic engineering, public transportation, electrified transportation, smart transportation, etc.</li> <li>2. To explain the operations of real-life transportation systems and the related engineering, economics, and environmental issues.</li> <li>3. To equip students with a broad knowledge base in transportation engineering, transportation electrification and sound practical sense, preparing them to choose the Transportation Systems Engineering degree path.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Intended Learning Outcomes</b>                     | <p>Upon completion of the subject, students will be able to:</p> <ol style="list-style-type: none"> <li>a. Identify the key issues in transportation systems, including electrified transportation systems.</li> <li>b. Appreciate the problems and suggest solutions to real-life transportation problems.</li> <li>c. Understanding the prospects and new trends in the transportation industry, including transportation electrification.</li> <li>d. Be ready to study subjects related to transportation, transportation electrification, and electric vehicles at a higher level.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Subject Synopsis/<br/>Indicative Syllabus</b>      | <ol style="list-style-type: none"> <li>1. <b>Transportation systems:</b> Introduction to transportation engineering, transportation problems and solutions, transportation systems in Hong Kong, transportation in social, economic, environmental, and political roles, and transportation modes.</li> <li>2. <b>Smart transportation:</b> Intelligent transportation systems, connected and autonomous vehicles, electrified transportation, shared mobility, low-altitude economy, and data-driven methods and solutions.</li> <li>3. <b>Traffic engineering:</b> Elements of traffic engineering, time-space diagram, speed-flow-density relationships, traffic measurement.</li> <li>4. <b>Public transportation (transit):</b> Designs, management, and operations of transit systems; transit network structures; service reliability and control; rail transportation.</li> <li>5. <b>Transportation systems analysis:</b> Transportation demand and supply; transportation policy, e.g. pricing; transportation network analysis.</li> </ol> |
| <b>Teaching/<br/>Learning<br/>Methodology</b>         | <p>The key concepts and techniques covered in this subject are discussed in lectures. Individual assignments consist of numerical problems, open-ended questions, and a final term essay, which let students demonstrate their level of understanding and create evidence of learning.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                            |             |                                                   |   |         |                               |          |  |  |  |   |   |   |   |          |   |   |   |   |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <table><tr><td rowspan="2">Learning/Learning Methodology</td><td colspan="4">Outcomes</td></tr><tr><td>a</td><td>b</td><td>c</td><td>d</td></tr><tr><td>Lectures</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr></table> |             |                                                   |   |         | Learning/Learning Methodology | Outcomes |  |  |  | a | b | c | d | Lectures | ✓ | ✓ | ✓ | ✓ |
| Learning/Learning Methodology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Outcomes                                                                                                                                                                                                                   |             |                                                   |   |         |                               |          |  |  |  |   |   |   |   |          |   |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | a                                                                                                                                                                                                                          | b           | c                                                 | d |         |                               |          |  |  |  |   |   |   |   |          |   |   |   |   |
| Lectures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ✓                                                                                                                                                                                                                          | ✓           | ✓                                                 | ✓ |         |                               |          |  |  |  |   |   |   |   |          |   |   |   |   |
| <b>Assessment Methods in Alignment with Intended Learning Outcomes</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                            |             |                                                   |   |         |                               |          |  |  |  |   |   |   |   |          |   |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Specific assessment methods/tasks                                                                                                                                                                                          | % weighting | Intended subject learning outcomes to be assessed |   |         |                               |          |  |  |  |   |   |   |   |          |   |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                            |             | a                                                 | b | c       | d                             |          |  |  |  |   |   |   |   |          |   |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1. Assignments                                                                                                                                                                                                             | 40%         | ✓                                                 | ✓ | ✓       | ✓                             |          |  |  |  |   |   |   |   |          |   |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2. Midterm exam                                                                                                                                                                                                            | 30%         | ✓                                                 | ✓ | ✓       | ✓                             |          |  |  |  |   |   |   |   |          |   |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 3. Final term essay                                                                                                                                                                                                        | 30%         | ✓                                                 | ✓ | ✓       | ✓                             |          |  |  |  |   |   |   |   |          |   |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Total                                                                                                                                                                                                                      | 100%        |                                                   |   |         |                               |          |  |  |  |   |   |   |   |          |   |   |   |   |
| <p>Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:</p> <p>The students will be assessed with three components: written assignments, a midterm examination, and a final term essay. The written assignments will consist of numerical, descriptive, real-system measurement, and diagnosis problems to address different aspects of skills required in achieving intended learning outcomes (a), (b), (c), and (d). The midterm exam is conducted near 2/3 of the semester (when most theoretical and quantitative contents have been taught) to consolidate students' knowledge in lectures, class activities, and self-learning activities. It appropriately assesses intended learning outcomes (a)-(d). The final term essay is conducted at the end of the semester to let students showcase and allow the lecturer to examine their knowledge learned about transportation systems, electrified transportation, and engineering practice, interests and contemplation on transportation issues, and motivations for choosing transportation engineering. It appropriately assesses intended learning outcomes (a)-(d).</p> |                                                                                                                                                                                                                            |             |                                                   |   |         |                               |          |  |  |  |   |   |   |   |          |   |   |   |   |
| <b>Student Study Effort Expected</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Class contact:                                                                                                                                                                                                             |             |                                                   |   |         |                               |          |  |  |  |   |   |   |   |          |   |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | ▪ Lectures                                                                                                                                                                                                                 |             |                                                   |   | 26 Hrs. |                               |          |  |  |  |   |   |   |   |          |   |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Other student study efforts:                                                                                                                                                                                               |             |                                                   |   |         |                               |          |  |  |  |   |   |   |   |          |   |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | ▪ Reading and studying                                                                                                                                                                                                     |             |                                                   |   | 26 Hrs. |                               |          |  |  |  |   |   |   |   |          |   |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | ▪ Completion of assignments                                                                                                                                                                                                |             |                                                   |   | 18 Hrs. |                               |          |  |  |  |   |   |   |   |          |   |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Total student study effort                                                                                                                                                                                                 |             |                                                   |   | 70 Hrs. |                               |          |  |  |  |   |   |   |   |          |   |   |   |   |

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| <b>Reading List and References</b> | <ol style="list-style-type: none"> <li>1. C.F. Daganzo, Fundamentals of Transportation and Traffic Operations, Pergamon, 2008.</li> <li>2. C.F. Daganzo and Y. Ouyang, Public Transportation Systems: Basic Principles of System Design, Operations Planning and Real-Time Control, 2019.</li> <li>3. J. Sussman, Introduction to Transportation Systems, Boston: Artech House, 2000.</li> <li>4. Jon D. Fricker and R.K. Whitford, Fundamentals of Transportation Engineering – A Multimodal Systems Approach. Prentice Hall, 2004.</li> <li>5. E. Quinet and R. Vickerman, Principles of Transport Economics, Edward Elgar Publishing Limited, 2004.</li> <li>6. A. Mohamed, A. Khan, A. Elsayed, and M. Elshaer, Transportation Electrification: Breakthroughs in Electrified Vehicles, Aircraft, Rolling Stock, and Watercraft, Wiley and IEEE Press Series on Power and Energy Systems, January 2023.</li> </ol> |
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July 2025